

Statistical Hypothesis Testing with R Examples

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1. Introduction

Statistical hypothesis testing is used to make inferences about population parameters based on sample data. We begin by formulating two competing hypotheses:

$$\begin{cases} H_0 : \text{Null hypothesis (no effect or no difference)} \\ H_1 : \text{Alternative hypothesis (there is an effect or difference)} \end{cases}$$

The decision rule is based on a test statistic, whose distribution under H_0 is known. We compute the *p-value* and compare it with a significance level α (commonly 0.05). If $p < \alpha$, we reject H_0 .

2. One-Sample *t*-Test

Model

We have a random sample $X_1, X_2, \dots, X_n$ from a normal population:

$$X_i \sim N(\mu, \sigma^2)$$

and we wish to test whether the population mean equals a specified value μ_0 .

Hypotheses

$$\begin{cases} H_0 : \mu = \mu_0 \\ H_1 : \mu \neq \mu_0 \quad (\text{two-tailed test}) \end{cases}$$

or alternatively, one-tailed tests can be formulated as:

$$H_1 : \mu > \mu_0 \quad \text{or} \quad H_1 : \mu < \mu_0$$

Test Statistic

The sample mean and sample variance are:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Then the test statistic is:

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$$

which follows a *t*-distribution with $n - 1$ degrees of freedom under H_0 .

Decision Rule

Reject H_0 if

$$|t| > t_{\alpha/2, n-1}$$

for a two-tailed test.

R Example

```
set.seed(123)
x <- rnorm(20, mean = 52, sd = 4)
t.test(x, mu = 50)
```

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3. Two Independent-Sample t -Test

Model

Let two independent random samples come from two populations:

$$X_1, X_2, \dots, X_{n_1} \sim N(\mu_1, \sigma_1^2), \quad Y_1, Y_2, \dots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$$

Hypotheses

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_1 : \mu_1 \neq \mu_2 \end{cases}$$

Case 1: Equal Population Variances ($\sigma_1^2 = \sigma_2^2$)

The pooled variance is:

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

and the test statistic is:

$$t = \frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with $n_1 + n_2 - 2$ degrees of freedom.

Case 2: Unequal Population Variances ($\sigma_1^2 \neq \sigma_2^2$)

The Welch's test statistic is:

$$t = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

The degrees of freedom are approximated using the Welch–Satterthwaite formula:

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1-1} + \frac{(S_2^2/n_2)^2}{n_2-1}}$$

R Example

```
set.seed(123)
group1 <- rnorm(25, mean = 10, sd = 3)
group2 <- rnorm(25, mean = 12, sd = 3)

# Equal variance assumption
t.test(group1, group2, var.equal = TRUE)

# Unequal variance (Welch's test)
t.test(group1, group2, var.equal = FALSE)
```

4. Paired t -Test

Model

For paired data, we compute differences:

$$D_i = X_i - Y_i, \quad i = 1, 2, \dots, n$$

and test whether the mean difference is zero.

Hypotheses

$$\begin{cases} H_0 : \mu_D = 0 \\ H_1 : \mu_D \neq 0 \end{cases}$$

Test Statistic

$$t = \frac{\bar{D}}{S_D / \sqrt{n}}$$

where

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i, \quad S_D^2 = \frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2$$

and $t \sim t_{n-1}$ under H_0 .

R Example

```
before <- c(80, 82, 78, 75, 89, 90)
after  <- c(85, 83, 80, 79, 88, 92)
t.test(before, after, paired = TRUE)
```

5. Test for Proportions

(a) One-Sample Proportion Test

Hypotheses

$$\begin{cases} H_0 : p = p_0 \\ H_1 : p \neq p_0 \end{cases}$$

Test Statistic

Let $\hat{p} = \frac{x}{n}$ be the sample proportion. Then:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

and under H_0 , $Z \sim N(0, 1)$.

R Example

```
prop.test(45, 100, p = 0.5, alternative = "two.sided", correct = FALSE)
```

—

(b) Two-Sample Proportion Test (Detailed Derivation and R Code)

Hypotheses

$$\begin{cases} H_0 : p_1 = p_2 \\ H_1 : p_1 \neq p_2 \end{cases}$$

Formulas

$$\hat{p}_1 = \frac{x_1}{n_1}, \quad \hat{p}_2 = \frac{x_2}{n_2}$$

Under H_0 , the combined (pooled) proportion is:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Then:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

and $Z \sim N(0, 1)$ under H_0 .

Detailed R Code

```
1 #####
2 # Two-sample proportion test in detail
3 # Example: prop.test(c(45, 55), c(100, 120), alternative = "two.
4 # sided", correct = FALSE)
5 #####
6
7 # 1. Input data
8 x1 <- 45; n1 <- 100
9 x2 <- 55; n2 <- 120
10
11 # 2. Sample proportions
12 p1_hat <- x1 / n1
13 p2_hat <- x2 / n2
14
15 cat("Sample proportions:\n")
16 cat("p1_hat =", round(p1_hat, 4), " p2_hat =", round(p2_hat, 4), "\n\n")
17
18 # 3. Null and alternative hypotheses
19 cat("H0: p1 = p2 vs. H1: p1 != p2 (two-sided test)\n\n")
20
21 # 4. Pooled proportion (under H0)
22 p_hat <- (x1 + x2) / (n1 + n2)
23 cat("Pooled proportion (under H0): p_hat =", round(p_hat, 4), "\n\n")
24
25 # 5. Standard error under H0
26 SE <- sqrt(p_hat * (1 - p_hat) * (1/n1 + 1/n2))
27 cat("Standard error (SE) =", round(SE, 5), "\n\n")
28
29 # 6. Test statistic (Z)
30 Z <- (p1_hat - p2_hat) / SE
31 cat("Test statistic (Z) =", round(Z, 3), "\n\n")
32
33 # 7. p-value
34 p_value <- 2 * (1 - pnorm(abs(Z))) # two-tailed p-value
35 cat("p-value =", round(p_value, 4), "\n\n")
36
37 # 8. 95% Confidence interval for (p1 - p2)
38 alpha <- 0.05
39 z_alpha2 <- qnorm(1 - alpha/2)
40 CI_lower <- (p1_hat - p2_hat) - z_alpha2 * SE
41 CI_upper <- (p1_hat - p2_hat) + z_alpha2 * SE
42 cat("95% CI for (p1 - p2): (", round(CI_lower, 4), ",", round(CI_upper, 4), ")\n\n")
43
44 # 9. Decision
45 if (p_value < alpha) {
46   cat("Result: Reject H0 at alpha =", alpha, "\n\n")
47 }
```

```

46 } else {
47   cat("Result: Fail to reject H0 (no significant difference)\n\n")
48 }
49
50 # 10. Compare with R's built-in test
51 cat("-----\n")
52 cat("Comparison with R's prop.test() output:\n\n")
53
54 result <- prop.test(c(x1, x2), c(n1, n2),
55                     alternative = "two.sided",
56                     correct = FALSE)
57 print(result)
58
59 # 11. Additional explanation
60 cat("\nExplanation:\n")
61 cat("- X-squared value in prop.test() equals Z^2.\n")
62 cat("- df = 1 because we compare two proportions.\n")
63 cat("- p-value close to 0.902 indicates no significant difference.\n")
64 cat("- Confidence interval includes zero, confirming equality of\n")
65   proportions.\n")
66 cat("- 'correct = FALSE' disables Yates continuity correction.\n")
67 cat("-----\n")

```

Interpretation

- The computed p-value (≈ 0.902) is large, so we fail to reject H_0 .
- This means there is no statistically significant difference between the two proportions.
- The 95% confidence interval includes zero, confirming the equality of proportions.
- In R's output, the X-squared value equals Z^2 , and the degrees of freedom are 1.

6. Summary Table

Test	Parameter	Test Statistic	Distribution
One-sample mean	μ	$\frac{\bar{X} - \mu_0}{S/\sqrt{n}}$	t_{n-1}
Two-sample mean (equal var.)	$\mu_1 - \mu_2$	$\frac{\bar{X} - \bar{Y}}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$	$t_{n_1+n_2-2}$
Paired samples	μ_D	$\frac{\bar{D}}{S_D/\sqrt{n}}$	t_{n-1}
One-sample proportion	p	$\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$	$N(0, 1)$
Two-sample proportion	$p_1 - p_2$	$\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$	$N(0, 1)$